🛒 CAPSTONE PROJECT REPORT

Project 1: Walmart Sales Forecasting

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a. Problem Statement

A retail store (Walmart) with multiple outlets across the country is facing significant challenges in managing its inventory. The core issue is the inability to accurately match product demand with supply, leading to either stock-outs or excess inventory. As a Data Scientist, the goal is to extract meaningful insights from historical data and build a predictive machine learning model to forecast future sales.

b. Project Objective

- To perform Exploratory Data Analysis (EDA) and discover hidden trends that impact weekly store sales.
- To evaluate multiple machine learning algorithms and choose the most robust model.
- To forecast the weekly sales for each store for the next **12 weeks** to optimize inventory and supply chain logistics.

c. Data Description

The dataset (Walmart.csv) consists of **6,435 rows** and **7 columns** (after text pre-processing). The features are detailed below:

- **Store:** The specific store number identifier.
- **Date:** The specific week of sales.
- **Weekly_Sales:** Total sales recorded for the given store in that week *(Target Variable)*.
- **Holiday_Flag:** Binary indicator (1 if it is a holiday week, 0 if it is a regular week).
- **month / year / week:** Structural time elements extracted from the date.

(Note: Features like Temperature, Fuel_Price, CPI, and Unemployment were analyzed and dropped during feature selection to avoid multi-collinearity and focus purely on time-based sales patterns).

d. Data Pre-processing Steps and Inspiration

1. **Handling Missing Values:** Checked the dataset using df.isnull().sum(). No missing values were found.
2. **Format Conversion:** The Date column was initially stored as text. It was converted into a standard datetime format (%Y-%m-%d) to ensure chronological order.
3. **Feature Engineering:** Extracted structural time elements (year, month, and week) from

the Date object. This was inspired by the need to capture seasonal trends like winter holiday shopping peaks.
4. **Data Splitting:** Divided the processed dataset into an **80% Training set** to teach the model and a **20% Testing set** to validate its real-world performance.

e. Choosing the Algorithm for the Project

Four different machine learning algorithms were tested and evaluated to find the best fit for this problem statement:

1. **Linear Regression** (Baseline model)
2. **Decision Tree Regressor**
3. **Random Forest Regressor**
4. **XGBoost Regressor** (Final Selected Model)

f. Motivation and Reasons For Choosing the Algorithm

- **Non-Linear Relationships:** Sales data is heavily affected by complex, non-linear factors (e.g., sudden spikes during holidays). **XGBoost (Extreme Gradient Boosting)** handles non-linear structures exceptionally well compared to linear models.
- **Boosting Technique:** XGBoost corrects errors sequentially from previous trees, leading to much lower errors and higher accuracy than individual decision trees.
- **High Efficiency:** It provides the highest R^2 Score and lowest Root Mean Squared Error (RMSE) on the validation set.

g. Assumptions

- It is assumed that historical sales trends, promotional holiday habits, and seasonal buying patterns will continue identically into the upcoming 12 weeks.
- Major external macroeconomic shocks (like a sudden pandemic or store closure) are assumed not to happen during the forecast period.

h. Model Evaluation and Techniques

The performance of the models was evaluated using standard regression evaluation metrics on the unseen 20% test dataset:

Model	R^2 Score (Accuracy)	RMSE (Lower is Better)
Linear Regression	0.12104	532,127
Random Forest	0.96630	104,128
Decision Tree	0.96874	100,347

Model	R^2 Score (Accuracy)	RMSE (Lower is Better)
XGBoost Regressor	**0.98013**	**80,004**

Conclusion: XGBoost Regressor outperformed all models with an accuracy (R^2 Score) of **98.01%** and a minimal error (RMSE) of **80,004**.

i. Inferences from the Same

- **Holiday Impact:** Store sales experience a massive surge during November and December (Months 11 & 12). This highlights the need for higher inventory planning before festivals like Thanksgiving and Christmas.
- **Store Variance:** Different stores show vastly different baseline sales averages. Inventory models must be customized per store rather than applied globally.
- **Yearly Decline:** Average sales showed a minor decreasing trend from 2010 to 2012, prompting the need for better marketing strategies.

j. Future Possibilities of the Project

- **Adding Store Metadata:** Integrating information about store sizes, specific product categories (e.g., clothing, electronics), and regional demographics would significantly improve forecast accuracy.
- **Promotion and Discount Data:** Incorporating historical markdowns (promotional discounts) would help predict exact sales lifts during promotional events.
